

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

MASTER IN COMPUTER APPLICATION (MCA)

Second Semester

Scheme of Examination



**DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

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SCHEME

SEMESTER -II

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem Exam	Tests	Assignment / Quiz	End Sem Practical Viva	Practical Record/Assignment/Quiz/Presentation	
1	MCA21	Operating System	3	1	-	4	70	20	10		-	100
2	MCA22	Information storage and Management	3	1	-	4	70	20	10	-	-	100
3	MCA23	Data Structures with C	3	1	-	4	70	20	10	-	-	100
4	MCA24	Computer oriented numerical and statistical Method	3	1	-	4	70	20	10	-	-	100
5	MCA25	Object-Oriented Programming With C++	3	1	-	4	70	20	10	-	-	100
6	MCA26	Object-Oriented Programming lab (C++)	-	-	8	8	-	-	-	100	50	150
7	MCA27	Data str. Lab	-	-	2	2	-	-	-	30	20	50
		Total	15	5	10	30	350	100	50	130	70	700

MCA-21 OPERATING SYSTEM

Unit-I

Introduction: Evolution of operating systems, Types of operating systems, Multitasking, Timesharing, Multithreading, Multiprogramming and, Real time operating systems, Different views of the operating system, System Programmer's view, User's view, Operating system concepts and structure, Layered Operating Systems, Monolithic Systems. Processes: The Process concept, The process control block, Operating system services for process management, Scheduling algorithms, First come first serve, Round Robin, Shortest run time next.

Unit-II

Memory Management : Memory management without swapping or paging, Concepts of swapping and paging, Page replacement algorithms namely, Least recently used, Optimal page replacement, Most recently used, Clock page replacement, First in First out (Stack algorithms), Modeling paging algorithms, Segmentation, Segmented Paging, Paged Segmentation.

Unit-III

Inter-process Communication and Synchronization: The need for inter-process synchronization, Concept of mutual exclusion, binary and counting semaphores, hardware support for mutual exclusion, queuing implementation of semaphores, Classical problems in concurrent programming, Dining Philosopher's problem, Bounded Buffer Problem, Sleeping Barber Problem, Readers and Writers problem, Critical section, critical region and conditional critical region, Monitors and messages.

Deadlocks: Concepts of deadlock detection, deadlock prevention, deadlock avoidance. Banker's Algorithm

Unit-IV

File System: File systems, directories, file system implementation, security protection mechanisms.

Input/output: Principles of I/O Hardware: I/O devices, device controllers, direct memory access. Principles of I/O software: Goals interrupt handlers, device drivers, and device independent I/O software. User space I/O Software.

Disks: Disk hardware, Disk scheduling algorithms (namely First come first serve, shortest seek time first algorithms) Error handling, track-at-a-time caching, RAM Disks.

Unit-V

Processes and Processors in Distributed Systems: Threads, System models, processor allocation, scheduling. Distributed File Systems: Design, Implementation, and trends. Performance Measurement, monitoring and evaluation Introduction, important trends affecting performance issues, why performance monitoring and evaluation are needed, performance measures, evaluation techniques.

Case Studies: WINDOWS and LINUX /UNIX Operating System.

BOOKS

1. Deitel, H.M. "An Introduction to Operating Systems". Addison Wesley Publishing Company.
2. Milenkovic, M., "Operating Systems - concepts and Design" McGraw Hill International Edition- Computer Science series 1992.
3. Galvin P., J.L. Abraham Silberschatz. "Operating System Concepts". John Wiley & Sons Company, 1989.
4. Tanenbaum, A.S. "Modern Operating System", Prentice Hall of India Pvt. Ltd.1995.
5. William Stallings "Operating Systems" , Prentice Hall of India Pvt. Ltd.
6. Joshi R.C. "Operating System" Wiley India.

MCA22-Information storage and Management

Unit-I

Introduction to Storage Technology: Data proliferation, evolution of various storage technologies, Overview of storage infrastructure components, Information Lifecycle Management, Data categorization.

Unit-II

Storage Systems Architecture: Intelligent disk subsystems overview, Contrast of integrated vs. modular arrays, Component architecture of intelligent disk subsystems, Disk physical structure Components, properties, performance, and specifications, RAID levels & parity algorithms, hot sparing, Front end to host storage provisioning, mapping and operation.

Unit-III

Introduction to Networked Storage: JBOD, DAS, NAS, SAN & CAS evolution and comparison. Applications, Elements, connectivity, standards, management, security and limitations of DAS, NAS, CAS & SAN.

Unit –IV

Hybrid Storage solutions; Virtualization: Memory, network, server, storage & appliances. Data center concepts & requirements, Backup & Disaster Recovery: Principles Managing & Monitoring: Industry management standards (SNMP, SMI-S, CIM), standard framework applications, Key management metrics (Thresholds, availability, capacity, security, performance).

Unit-V

Information storage on cloud :Concept of Cloud, Cloud Computing, storage on Cloud, Cloud Vocabulary, Architectural Framework, Cloud benefits, Cloud computing Evolution, Applications & services on cloud, Cloud service providers and Models, Essential characteristics of cloud computing, Cloud Security and integration.

BOOKS

1. G. Somasundaram & Alok Shrivastava (EMC Education Services) editors; Information Storage and Management: Storing, Managing, and Protecting Digital Information; Wiley India.
2. Ulf Troppens, Wolfgang Mueller-Friedt, Rainer Erkens, Rainer Wolafka, Nils Haustein; Storage Network explained : Basic and application of fiber channels, SAN, NAS, iSES, INFINIBAND and FCOE, Wiley India.
3. John W. Rittinghouse and James F. Ransome; Cloud Computing : Implementation , Management and Security, CRC Press, Taylor Frances Pub.
4. Nick Antonopoulos, Lee Gillam; Cloud Computing : Principles, System & Application, Springer.
5. Anthony T. Velete, Toby J.Velk, and Robert Eltenpeter, Cloud Computing : A practical Approach, TMH Pub.

MCA23-Data Structures with C

UNIT-I

Stack and Queue: contiguous implementations of stack, various operations on stack, various polish notations-infix, prefix, postfix, conversion from one to another-using stack; evaluation of post and prefix expressions. Contiguous implementation of queue: Linear queue, its drawback; circular queue; various operations on queue; linked implementation of stack and queue-operations

UNIT-II

General List: list and it's contiguous implementation, it's drawback; singly linked list-operations on it; doubly linked list-operations on it; circular linked list; linked list using arrays.

UNIT-III

Trees: definitions-height, depth, order, degree, parent and child relationship etc; Binary Trees-various theorems, complete binary tree, almost complete binary tree; Tree traversals-preorder, in order and post order traversals, their recursive and non recursive implementations; expression tree- evaluation; linked representation of binary tree-operations.

UNIT-IV

Searching, Hashing and Sorting: requirements of a search algorithm; sequential search, binary search, indexed sequential search, interpolation search; hashing-basics, methods, collision, resolution of collision, chaining; Internal sorting- Bubble sort, selection sort, insertion sort, quick sort, merge sort on linked and contiguous list, shell sort, heap sort, tree sort.

UNIT-V

Graphs: related definitions: graph representations- adjacency matrix, adjacency lists, adjacency multilist; traversal schemes- depth first search, breadth first search; Minimum spanning tree; shortest path algorithm; kruskals & dijkstras algorithm. Miscellaneous features Basic idea of AVL tree- definition, insertion & deletion operations; basic idea of B-tree- definition, order, degree, insertion & deletion operations; B+-Tree- definitions, comparison with B-tree; basic idea of string processing.

BOOKS

1. Aho "Data Structure & Algorithms".
2. Yashwant Kanetkar, Understanding Pointers in C, BPB.
3. Tennenbaum A.M. & others: Data Structures using C & C++; PHI
4. Horowitz & Sawhaney: Fundamentals of Data Structures, Galgotia Publishers.
5. Kruse R.L. Data Structures and Program Design in C; PHI
6. Trembly "Introduction to Data Structure with Applications".

MCA24-Computer oriented numerical and statistical Method

UNIT – I

Numerical approximation, Representation of integers and real numbers in computers, fixed and floating point arithmetic, normalized floating point numbers, Round off and truncation errors, relative and absolute errors. Newton Raphson methods. Convergence of solutions.

Unit – II

Interpolation : Forward, Backward, central (Stirling's) and divided difference formulas, Lagrange's interpolation, Inverse interpolation for equal and unequal intervals.

Numerical Integration : Newton Cotes's formula, Simpson's 1/3rd and 3/8th rule. Gauss Legendre (two and three points) integration formula.

Unit – III

Simultaneous linear equations: Solutions of simultaneous linear equations – Gauss elimination method and pivoting, ill conditioned equations and refinement of solutions, Gauss-Seidel iterative methods. Solution of differential equation: Runge-Kutta fourth order method. Euler's method, Picard's, Taylor's series.

Unit – IV

Distributions: Binomial distribution, Poisson distribution and normal distribution, χ^2 distribution, Rectangular distribution, hypergeometric distribution.

Unit – V

Hypothesis testing for sampling: Small samples, t, z and f tests. Chi-square test.

Large samples : Comparison of large samples, testing the significance of the difference between the means of two large samples.

BOOKS

1. E. Balaguruswamy "Numerical Methods" , TMH, ISBN – 07-463311-2, 1999.
2. B.S. Grewal "Numerical Methods in Engineering & Science".
3. E. Balaguruswamy "Numerical Methods" , TMH, ISBN – 07-463311-2, 1999.
4. V. Rajaraman "Computer Oriented Numerical Methods".
5. M. Ray and Har Swarup Sharma "Mathematical Statistics".

MCA25-Object-Oriented Programming with C++

UNIT –I

Overview of C++: Object oriented programming, Concepts, Advantages, Usage. C++ Environment: Program development environment, the language and the C++ language standards., loops and decisions, structures and functions, object and classes, object arrays, construct and destructor functions

UNIT-II

Array, Pointers References & The Dynamic Allocation Operators: Array of Objects, Pointers to Object, Type Checking C++ Pointers, The This Pointer, Pointer to Derived Types, Pointer to Class Members, References: Reference Parameter, call by reference and return by reference Passing References to Objects, Operator and function overloading, pointers, pointers to base and derived classes inheritance, public and private inheritance, multiple inheritance.

UNIT-III

Polymorphism, virtual functions, abstract base classes and pure virtual function, friend function, early and late binding.

UNIT-IV

C++ I/O system, formatted I/O, creating insertors and extractors, file I/O basis, creating disk files and file manipulations using seekg(), seekp(), tellg() and tellp() functions, exception handling: try, catch and throw.

UNIT-V

UML concepts, object-oriented paradigm and visual modeling, UML diagrams, UML specifications, object model, object oriented design, identifying classes and object, object diagrams.

BOOKS

1. Hans Erit Eriksson “UML 2 toolkit” Wiley
2. Lee “UML & C++ a practical guide to Object Oriented Development 2 ed, Pearson.
3. Schildt “C++ the complete reference 4ed, 2003.
4. Hans Erit Eriksson “UML 2 toolkit” Wiley.
5. Balagurusawmy “Object Orienter Programming with C++”.
6. B.G., Boach “Object Oriented Analysis & Design with Applications”, Addison Wesley.
7. S. Parate “C++ Programming”, BPB

